

A Non-Fin Obstruction for $c_{0,\mathcal{I}}\hat{\otimes}_\varepsilon X$

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Status

Candidate partial result, likely valid. This packet gives a sharp answer to Question 5.12 of Rincon-Villamizar and Uzcategui Aylwin [1] for all Banach spaces X containing no complemented copy of c_0 , and in particular for $X = \ell_2$.

Source Question

The source paper proves that for every Banach space X and ideal $\mathcal{I}$ on $\mathbb{N}$, the canonical map

$$\Phi_{\mathcal{I},X} : c_{0,\mathcal{I}}\hat{\otimes}_\varepsilon X \longrightarrow c_{0,\mathcal{I}}(X), \quad \sum_{j=1}^m a^j \otimes x_j \mapsto \left(\sum_{j=1}^m a_n^j x_j \right)_{n \in \mathbb{N}},$$

extends to an isometry into $c_{0,\mathcal{I}}(X)$. The paper then asks:

Remark 5.11. The above isometry is not onto in general. Indeed, if $\mathcal{I} = \mathcal{P}(\mathbb{N})$, then $c_{0,\mathcal{I}}$ is ℓ_∞ and it is known that the existence of an isometry from $\ell_\infty \bigvee X$ onto $\ell_\infty(X)$ implies that X contains a complemented copy of c_0 (see [20]). However, when $\mathcal{I} = \text{Fin}$, the above isometry is onto as it is established in [7, Theorem 1.1.11].

Question 5.12. For which ideals $\mathcal{I}$, is $c_{0,\mathcal{I}} \bigvee X$ isometric to $c_{0,\mathcal{I}}(X)$?

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This packet reads Question 5.12 in the context of Remark 5.11: when is the above canonical isometry onto?

Proof Intuition

The obstruction already appears in the source paper for the largest ideal $\mathcal{P}(\mathbb{N})$: onto-ness of $\ell_\infty \hat{\otimes}_\varepsilon X \rightarrow \ell_\infty(X)$ forces X to contain a complemented copy of c_0 . Any proper ideal $\mathcal{I} \neq \text{Fin}$ contains an infinite set A . On the coordinates in A , the restriction ideal is the full powerset $\mathcal{P}(A)$, so $c_{0,\mathcal{I}}$ contains a coordinate $\ell_\infty(A)$ block. If the global tensor map were onto, projecting onto that block would make the $\ell_\infty(A)$ tensor map onto. Thus the same Leung–Rabiger obstruction rules out every non-Fin ideal whenever X has no complemented c_0 .

Restriction Diagram

Lemma 1. *Let $\mathcal{I}$ be an ideal on $\mathbb{N}$, let $A \subseteq \mathbb{N}$, and let X be a Banach space. Suppose the canonical map*

$$\Phi_{\mathcal{I},X} : c_{0,\mathcal{I}} \hat{\otimes}_\varepsilon X \rightarrow c_{0,\mathcal{I}}(X)$$

is onto. Then the canonical map

$$\Phi_{\mathcal{I}|A,X} : c_{0,\mathcal{I}|A} \hat{\otimes}_\varepsilon X \rightarrow c_{0,\mathcal{I}|A}(X)$$

is onto as well.

Proof. Identify $c_{0,\mathcal{I}|A}$ with the coordinate subspace of $c_{0,\mathcal{I}}$ supported on A . The coordinate projection $P_A : c_{0,\mathcal{I}} \rightarrow c_{0,\mathcal{I}}$, given by $(P_A a)_n = a_n$ for $n \in A$ and 0 otherwise, maps $c_{0,\mathcal{I}}$ into itself and has range $c_{0,\mathcal{I}|A}$. Indeed, for every $\delta > 0$,

$$\{n : |(P_A a)_n| \geq \delta\} = A \cap \{n : |a_n| \geq \delta\} \in \mathcal{I}.$$

Similarly, the coordinate projection $Q_A : c_{0,\mathcal{I}}(X) \rightarrow c_{0,\mathcal{I}|A}(X)$ has range $c_{0,\mathcal{I}|A}(X)$.

Let $y \in c_{0,\mathcal{I}|A}(X)$, viewed as an element of $c_{0,\mathcal{I}}(X)$ supported on A . Since $\Phi_{\mathcal{I},X}$ is onto, choose $u \in c_{0,\mathcal{I}} \hat{\otimes}_\varepsilon X$ with $\Phi_{\mathcal{I},X}(u) = y$. The injective tensor product is functorial for bounded operators, so $(P_A \otimes I_X)u \in c_{0,\mathcal{I}|A} \hat{\otimes}_\varepsilon X$, and by coordinatewise commutativity,

$$\Phi_{\mathcal{I}|A,X}((P_A \otimes I_X)u) = Q_A \Phi_{\mathcal{I},X}(u) = Q_A y = y.$$

Thus $\Phi_{\mathcal{I}|A,X}$ is onto. □

Main Partial Result

Theorem 1. *Let X be a Banach space containing no complemented copy of c_0 . For an ideal $\mathcal{I}$ on $\mathbb{N}$ containing all finite sets, the canonical isometry*

$$\Phi_{\mathcal{I},X} : c_{0,\mathcal{I}} \hat{\otimes}_\varepsilon X \rightarrow c_{0,\mathcal{I}}(X)$$

is onto if and only if $\mathcal{I} = \text{Fin}$.

Proof. If $\mathcal{I} = \text{Fin}$, onto-ness is the classical identity

$$c_0 \hat{\otimes}_\varepsilon X \cong c_0(X),$$

quoted in the source paper from [2, Theorem 1.1.11].

Conversely, suppose $\mathcal{I} \neq \text{Fin}$. Then there is an infinite set $A \in \mathcal{I}$. Since $\mathcal{I}$ is hereditary, every subset of A belongs to $\mathcal{I}|A$, so $\mathcal{I}|A = \mathcal{P}(A)$. Hence

$$c_{0,\mathcal{I}|A} = \ell_\infty(A), \quad c_{0,\mathcal{I}|A}(X) = \ell_\infty(A, X).$$

If $\Phi_{\mathcal{I},X}$ were onto, the restriction lemma would imply that

$$\ell_\infty(A) \hat{\otimes}_\varepsilon X \rightarrow \ell_\infty(A, X)$$

is onto. Since A is countably infinite, this is the same as the canonical isometry $\ell_\infty \hat{\otimes}_\varepsilon X \rightarrow \ell_\infty(X)$ being onto. By the Leung–Rabiger result quoted in Remark 5.11 of [1], this forces X to contain a complemented copy of c_0 , contrary to the hypothesis. Therefore $\mathcal{I} = \text{Fin}$. □

Corollary 1. *For $X = \ell_2$, the canonical map*

$$c_{0,\mathcal{I}} \hat{\otimes}_\varepsilon \ell_2 \rightarrow c_{0,\mathcal{I}}(\ell_2)$$

is onto exactly for $\mathcal{I} = \text{Fin}$.

Proof. The Hilbert space ℓ_2 contains no copy of c_0 , complemented or otherwise. □

Verification Notes

No computation is used. The proof depends on:

- the coordinate restriction projection on $c_{0,\mathcal{I}}$ and on $c_{0,\mathcal{I}}(X)$;
- functoriality of the injective tensor product under the projection P_A ;
- the source paper's quoted Leung–Rabiger obstruction for $\ell_\infty \hat{\otimes}_\varepsilon X \rightarrow \ell_\infty(X)$.

The result is intentionally phrased for the source's canonical isometry being onto. It does not claim a classification of arbitrary Banach-space isometric isomorphisms between $c_{0,\mathcal{I}} \hat{\otimes}_\varepsilon X$ and $c_{0,\mathcal{I}}(X)$.

Novelty Check

Local run-index searches for 2309.08076, `injective tensor`, `c_{0,\mathcal{I}}`, `Leung`, and `c_{0,\mathcal{I}}(X)` found no prior packet or attempt for this tensor-product target. The only exact arXiv-id hit was the separate $c_{0,\text{WO}}$ Grothendieck counterexample packet.

Bounded web searches on 2026-06-29 for exact and close phrases from Question 5.12, including

`"c_{0,\mathcal{I}}" "injective tensor product" "c_{0,\mathcal{I}}(X)"`

and the source title with `injective tensor`, found no explicit later answer in this pass.

Limitations

This is a partial answer to Question 5.12. It fully classifies the ideals for each X with no complemented copy of c_0 , but it leaves open the structure of non-Fin ideals for spaces X that do contain complemented copies of c_0 .

Human Review Recommendation

Likely valid. Review should focus on the reading of Question 5.12 as asking about the source's canonical isometry and on the restriction lemma.

References

- [1] M. A. Rincon-Villamizar and C. Uzcategui Aylwin, *Banach spaces of $\mathcal{I}$ -convergent sequences*, arXiv:2309.08076, 2023.
- [2] J. Diestel, J. H. Fourie, and J. Swart, *The metric theory of tensor products. Grothendieck's resume revisited*, American Mathematical Society, Providence, RI, 2008.
- [3] D. Leung and F. Rabiger, *Complemented copies of c_0 in ℓ_∞ sums of Banach spaces*, Illinois J. Math. 34 (1990), 52–58.